even though the correct grade is in the engine;

- Diesel dilution of the oil—this can occur on engines with internal fuel lines. The oil level will rise; once enough diesel has found its way into the oil to lower the pressure to a noticeable extent, it will be possible to smell the fuel in the oil if you take a sample from the dipstick and rub it on your fingers.
- Worn bearings. The oil pump feeds oil under pressure through holes drilled in the crankcase (called *galleries*) and through various pipes to all the engine bearings. Oil squeezes out between the two surfaces of the bearings. The pump has a large enough capacity to circulate more oil than is needed to maintain the correct pressure in the system, and the surplus bleeds back to the sump via a *pressure relief valve*. In time, wear on the bearings allows the oil to flow out more freely. Eventually the rate of flow is greater than the pump can sustain at the normal operating pressure, and the pressure declines.

  Worn bearings do not, as a rule, develop overnight. A very gradual decline in oil pressure occurs, especially at low engine speeds when the engine is hot. There will also be a gradual increase in engine noise and knocks. By the time a significant loss of oil pressure occurs, many other problems are likely to be evident, and a major rebuild is called for. *Any rapid loss of oil pressure accompanied by a new engine knock indicates a specific bearing failure that needs immediate attention.*
- Oil pressure relief valves sometimes malfunction, generally through a broken spring or because of trash on the valve seat (especially if proper oil change procedures have been ignored). In this case, the oil is vented directly back to the sump, with a consequent loss in pressure.

  Problems with pressure relief valves are rare but are simple to check. Almost invariably, the pressure relief valve is screwed into the side of the block somewhere or fitted to the oil filter mount and can be easily removed, disassembled, cleaned, and put back. The spring is liable to be under some tension, so take care when taking the valve apart. After cleaning, reset the spring's tension to maintain the manufacturer's specified oil pressure. Run the engine until it is warm and check the oil pressure. If it is low, shut down the engine and tighten the relief valve spring a little or add more shims. If no amount of tensioning on the spring brings the oil pressure up to the manufacturer's specifications, the problem lies elsewhere.
- Oil pumps rarely, if ever, give out, as long as the oil is kept topped up and clean and the filter is changed regularly. Over a long period of time, wear in an oil pump may produce a decline in pressure, but not before wear in the rest of the engine creates the need for a major rebuild. At this time, the oil pump should always be checked.
- A clogged oil cooler will reduce oil pressure. Clogging can arise from long hours of low load running and idling (which generates a carbon sludge within the engine), coupled with a failure to change the oil at the specified intervals (or using non-detergent oil in the engine). In this case, other oil galleries are likely to be plugged with sludge and serious engine damage is probable. The oil side of most oil coolers cannot be flushed—the oil cooler will need replacing.
- A well-heeled sailboat will sometimes cause the oil pump suction line to come clear of the oil in the pan (sump), allowing the pump to suck in a slug of air. The oil pressure will momentarily drop, generally with a sudden, alarming clatter from the engine. This is especially likely to happen if the oil level is a little low. Check the level, top up as necessary, and put the boat on a more even keel.
- The failure of an external oil line or gasket (e.g., to an oil cooler) will cause a sudden, potentially catastrophic loss of oil and pressure. The engine is likely to suddenly clatter loudly. It must be shut down immediately. There will be oil all over the engine room! Less easy to spot is the loss of oil that accompanies a corroded cooling tube in an oil cooler. The oil will be pumped out of the exhaust (or into the header tank), and sometimes water will enter the crankcase (see above).